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Section A

1. Database

= D. It's a place where data is stored and while it's here it can be altered or deleted.

- Database is important because in today's world on 4th Industrial, everything is data and this data needs a ~~place~~ place where it can be stored and safely and it should be ~~able~~ able to be accessed quickly.

Example :

Bright learn could make a database where it contains students name, course, marks and fees, instead of keeping hundreds of students records in paper files.

2. Data Warehouse

= This is where multiple database is stored and write on the database, here you just store data you cannot manipulate it

- On a regular database you should be able to manipulate data like retrieve and delete it but on a warehouse you cannot, you just store it.

3. Data Lake

= Data lakes stores original data before it's manipulated or cleaned and on a data warehouse it has to be cleaned and structured or organised

= A company would prefer Data Lake if it has a lot of unstructured data and wishes to use it in the future.

4. Data Mart

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5. - Structured Data is organised data, organised in rows and columns
e.g.

- Semi Structured Data is data that has been half organised, it has some data organised but no set tables
e.g. An xml file

- Unstructured Data is data that has no set format
e.g. music, videos, photos

6. Data Modelling is when we are planning on how to organise data and how to piece it together.

- It's important because a good plan can help the system just

Two types : ① A conceptual model
② A logical model

7. Data Mining is going through a lot of data trying to find useful patterns - businesses use it to understand customers and which days of the month are ~~less~~ more busy than others

8. A DBMS is a system that lets you create, manage and use databases

- MySQL = Its free and mostly used for websites and small apps

- PostgreSQL = Also free and good for complex data and analytics

- Oracle = used by banks

- MongoDB = good for apps where data changes often

9. ETL: E = Extract = collect data from database and files
T - Transform = clean it and change it.
L - Load = put it into warehouse

ETL moves data from many places into one place.

- It matters because to analyse data analysts need clean and consistent data.

10. Storage formats

- CSV = simple texts and opens in Excel

- JSON = texts formats and use for web APIs

- Parquet = compressed format and stores data by column

- Avro = a compact format that stores data by row

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